

Project Report: Timer Controlled Relay PCB Design

Electronic System Design - I Project

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Abstract

This report presents the design, theoretical operation, and layout procedure for a Timer Controlled Relay PCB implemented using KiCad EDA. The circuit utilizes an NE555D precision timing IC paired with an MPSA42 switching transistor to drive an electromechanical SPDT relay. The board is constructed using a 4-layer FR4 PCB stackup featuring reverse-polarity input protection, inductive flyback snubbing, and visual LED status monitoring. Detailed analysis of the descending-voltage RC timing topology, cold-start wiring variants, and hardware design improvements over off-the-shelf commercial timer modules are thoroughly documented.

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1 Introduction to the Project Work

The **Timer Controlled Relay** is a power-switching interface designed to automatically actuate secondary electrical loads after a defined, adjustable time delay.

Electromechanical relays allow low-power control circuits to safely switch high-current or high-voltage devices. Incorporating an onboard precision timer removes the need for continuous microcontroller control, making this design ideal for standalone automation tasks, safety cut-offs, power line settling, and delayed motor starting.

1.1 Project Specifications

- **Timing IC:** NE555D precision timer (SOIC-8 package).
- **Switching Element:** Single Pole Double Throw (SPDT) Relay driven by an MPSA42 NPN BJT.
- **Input Protection:** 1N5819 Schottky diode for reverse-polarity protection.
- **PCB Architecture:** 4-Layer FR4 copper stackup (exceeding minimum double-layer requirement).
- **User Interface:** Precision trimmer potentiometer (RV_2) for delay tuning, with power and relay activation LEDs.

2 Detailed Description & Theoretical Background

2.1 NE555 On-Delay Timing Theory (Descending RC Curve)

The core timing mechanism operates on a descending-voltage RC topology [1]. Trigger pin (Pin 2) and Threshold pin (Pin 6) are connected together to form a single sense node tied between timing capacitor C_3 and trimmer potentiometer RV_2 [1].

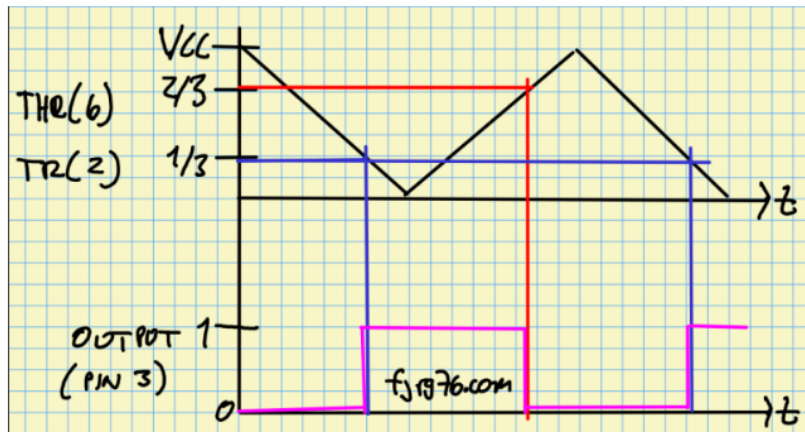


Figure 1: 555 On-Delay Waveform: Pins 2 & 6 voltage decay vs Output state (Pin 3) [1].

The timing sequence unfolds as follows:

1. **Power-On ($t = 0$):** A discharged capacitor acts as a momentary short circuit [1]. Because C_3 is located at the top of the RC network connected to V_{CC} , the node voltage at Pins 2 and 6 instantly jumps to V_{CC} [1]. Because this voltage exceeds $\frac{2}{3}V_{CC}$, the 555 internal flip-flop holds Output (Pin 3) **LOW (0V)***, keeping the relay deactivated [1].

2. **Capacitor Charging Phase:** As C_3 charges through RV_2 , the potential at the $(-)$ terminal node slowly decreases, exponentially approaching 0V [1].
3. **Threshold Trip & Relay Activation:** When the node potential drops below $\frac{1}{3}V_{CC}$, the internal comparator triggers, causing Output (Pin 3) to transition ****HIGH (V_{CC})**** [1]. This energizes relay coil K_1 , closing the contacts and activating the load [1].

The duration T required for the node voltage to drop from V_{CC} down to the $\frac{1}{3}V_{CC}$ threshold is governed by Equation 1:

$$T = 1.1 \times (R_4 + RV_2) \times C_3 \quad (1)$$

2.2 Transistor Switching & Saturation

Because the NE555 output pin (Pin 3) drives the relay coil, an MPSA42 NPN BJT (Q_2) is configured as a low-side switch. To drive the transistor into full saturation ($V_{CE(sat)} \approx 0.2V$), the base current I_B must satisfy:

$$I_B \geq \frac{I_C}{\beta_{min}} = \frac{I_{coil}}{\beta_{min}} \quad (2)$$

Resistor R_5 limits the base current to prevent over-driving the transistor while ensuring saturation upon activation.

2.3 Inductive Flyback Suppression

When current through the relay coil K_1 is cut off, the collapsing magnetic field creates a reverse voltage spike according to $V = -L \frac{di}{dt}$. Flyback diode D_7 (1N4148) connected across the coil provides a recirculation path for current, protecting transistor Q_2 from high-voltage breakdown.

2.4 Cold-Start Prevention Wiring Variant

In power-sensitive applications where downstream circuits must be protected from initial V_{CC} power-up transients (cold starts), the relay output terminals are wired as follows [1]:

- **Supply Connection:** Main supply voltage V_{CC} is wired directly to the Common (C) terminal of the relay block [1].
- **Load Output Connection:** The positive power terminal of the secondary circuit is connected to the Normally Open (NO) terminal [1].
- **Operation:** Power is completely disconnected from the load during initial boot and is only applied once the 555 timer completes its delay cycle and closes the C -to- NO contacts [1].

3 Step-by-Step Design Procedure

3.1 Step 1: Schematic Capture in KiCad Eeschema

The complete circuit diagram was captured in KiCad Eeschema. Components were placed, assigned nets, and checked via Electrical Rules Check (ERC).

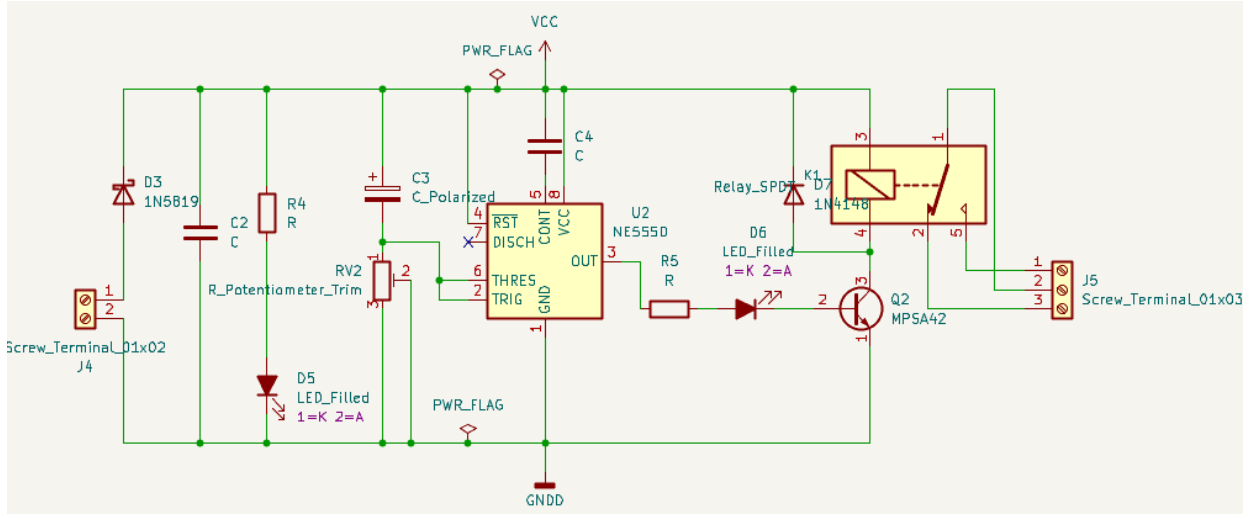
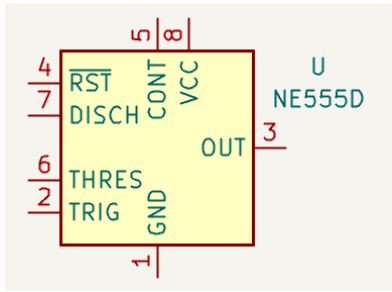
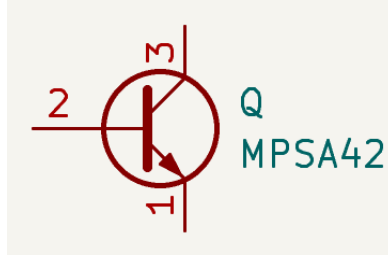


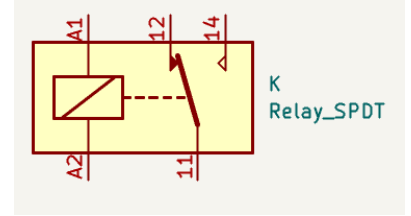
Figure 2: Complete Schematic Capture in Eeschema.



(a) NE555D IC Symbol



(b) MP5A42 Transistor Symbol

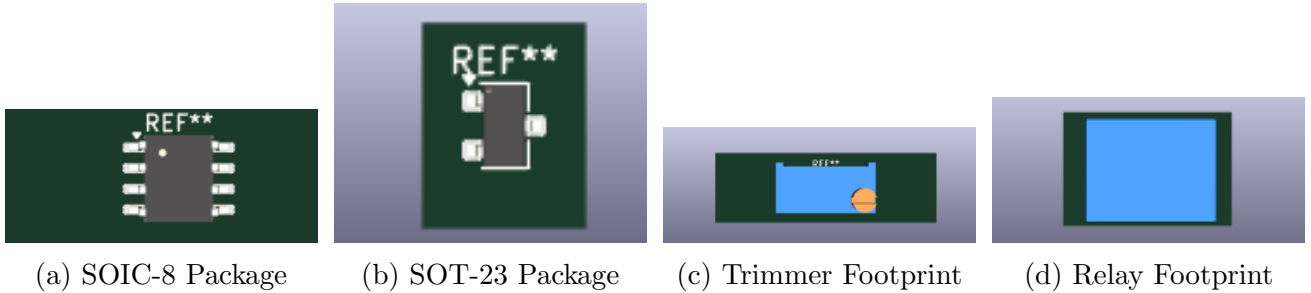


(c) SPDT Relay Symbol

Figure 3: Key Active Schematic Symbols.

3.2 Step 2: Footprint Mapping and Component Selection

Each schematic symbol was mapped to physical footprints in the KiCad footprint library. A mixture of Surface-Mount Devices (SMD) and Through-Hole (THT) components was utilized.



(a) SOIC-8 Package

(b) SOT-23 Package

(c) Trimmer Footprint

(d) Relay Footprint

Figure 4: Physical Footprint Libraries Mapped in KiCad.

3.3 Step 3: Board Setup & 4-Layer Physical Stackup

A 4-layer FR4 copper stackup was configured in KiCad Board Setup. Internal layers provide dedicated ground planes and power distribution paths, ensuring signal isolation and low noise.

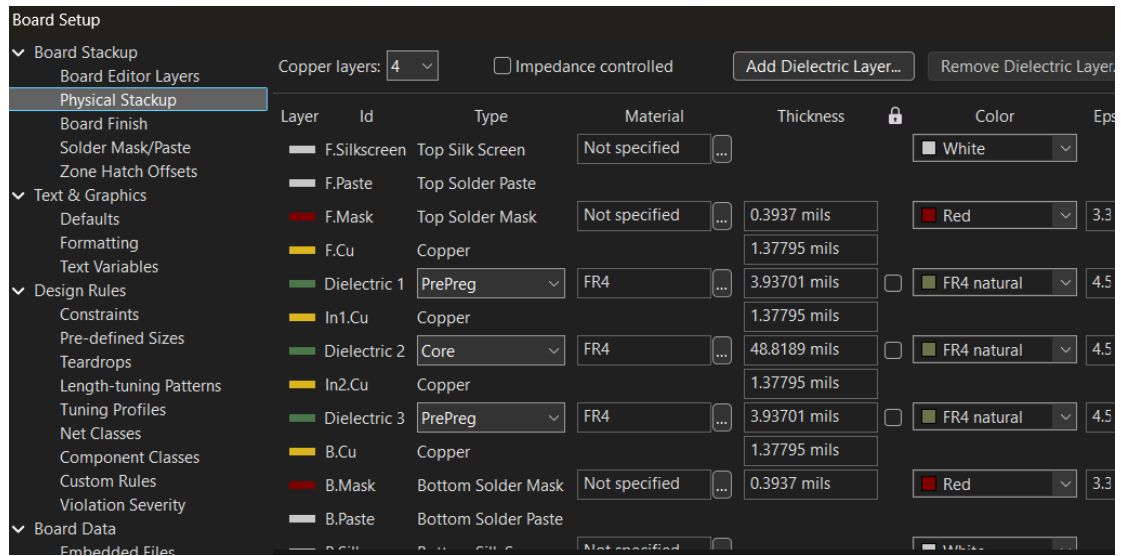
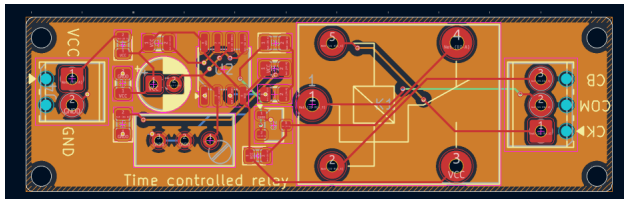


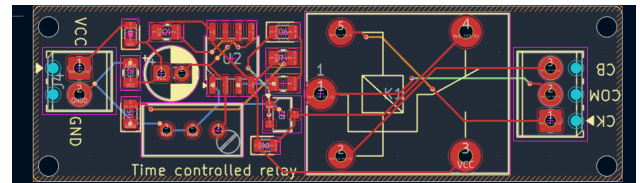
Figure 5: 4-Layer Copper Board Stackup Configuration.

3.4 Step 4: Component Placement & PCB Routing

Components were arranged logically to keep low-power timing traces separate from high-current relay tracks. Traces were routed, copper ground zones filled, and stitching vias inserted.



(a) Top Layer & Copper Zones

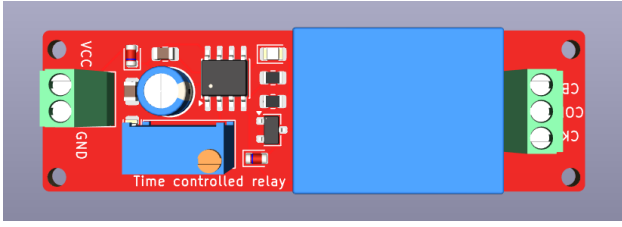


(b) Trace Routing Connections

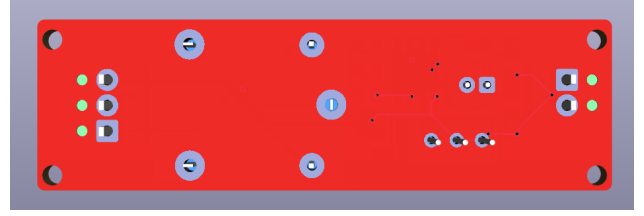
Figure 6: PCB Layout Execution in KiCad PCB Editor.

3.5 Step 5: Design Rule Check (DRC) & 3D Visualization

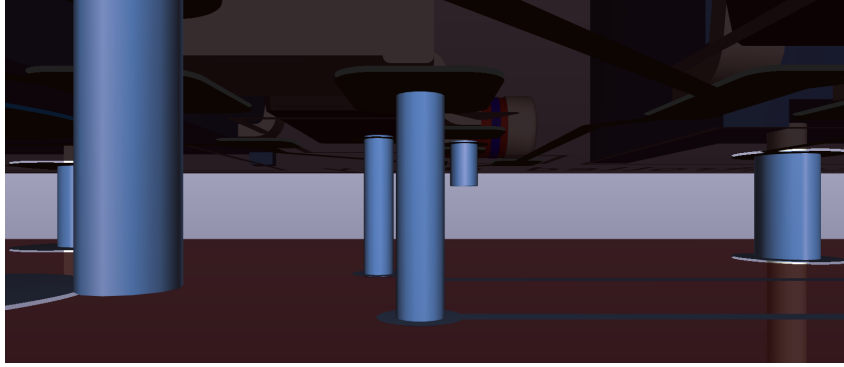
KiCad's 3D Ray-Tracing Engine was utilized to inspect physical clearances, part orientations, connector accessibility, and pin solder pad alignments.



(a) Top Assembled View



(b) Bottom Layer View



(c) Under-Board Pin Clearance Perspective

Figure 7: Photorealistic 3D Inspection Views.

4 Bill of Materials (BOM)

Table 1: Project Bill of Materials

Ref	Qty	Value	Package	Description
U_2	1	NE555D	SOIC-8	Precision Timer IC
Q_2	1	MPSA42	SOT-23	NPN Switching Transistor
K_1	1	Relay_SPDT	SPDT Module	Power Switch Actuator
D_3	1	1N5819	SOD-123	Reverse Polarity Schottky Diode
D_7	1	1N4148	SOD-123	Flyback Protection Diode
D_5, D_6	2	LED_Filled	0805 SMD	Status Indicators
RV_2	1	Potentiometer	Trimmer	Adjustable Timing Control
C_3	1	Polarized Cap	Radial THT	RC Timing Storage Capacitor
C_2, C_4	2	Ceramic Cap	0805 SMD	Power Decoupling Capacitors
J_4, J_5	2	Screw Terminals	2-Pin / 3-Pin	Power In / Load Contacts

5 Potential Applications, Usability Enhancements & Commercial Design Comparison

5.1 Potential Application Areas

1. **Industrial Motor Soft-Start & Power Sequencing:** Delays secondary heavy-load activation to mitigate inrush current spikes and prevent main supply brownouts.
2. **Cold-Start Protection for Sensitive Electronics:** Holds off power delivery until raw supply voltages stabilize to nominal operating levels [1].

3. **Automotive After-Run Timers:** Keeps cooling fans or auxiliary pumps active for a preset duration after engine ignition turn-off.
4. **Smart Home Safety Cut-Offs:** Implements an automatic fail-safe turn-off for high-power heating elements or pumps left unattended.

5.2 Usability Enhancements & Hardware Modifications

- **Optocoupler Isolation (Galvanic Safety):** Inserting an optocoupler (e.g., PC817) between the 555 output and the transistor base isolates high-voltage AC load transients from low-voltage DC logic ground.
- **Microcontroller Interface Header:** Adding a 3-pin breakout header (VCC, GND, TRIG) enables direct external control via Arduino, ESP32, or Raspberry Pi.
- **Wide-Voltage Input LDO Regulator:** Integrating an onboard LDO regulator (e.g., LM7812) allows operation across supply voltages from 7V to 30V DC without modifying coil drive circuitry.
- **Digital OLED Display & Encoder:** Replacing the analog potentiometer with a digital rotary encoder and I2C OLED display enables precise delay calibration without trial-and-error adjustment.

5.3 Comparative Analysis: Flaws in Standard Commercial Timer Boards

Mass-produced commercial timer modules (often referred to as low-cost commercial boards) contain severe layout and electrical engineering shortcomings [1]:

- **Tuning Inconvenience:** They utilize uncalibrated multiturn potentiometers with no visual indicators, forcing user reliance on time-consuming trial and error [1].
- **Poor Thermal Relief Serviceability:** Trimmer terminals are frequently soldered directly into unbroken bottom ground planes without thermal relief, making rework or replacement extremely difficult [1].
- **High-Voltage Safety Risks:** Commercial boards lack physical isolation slots (air gaps) between the high-voltage Common terminal and the relay coil pins/ground planes, inviting electrical arcing and PCB charring [1].
- **Inadequate Current Carrying Capacity:** Output traces are routing-constrained and too narrow for 10A–15A loads; solder mask application prevents manual reinforcement with extra solder tin [1].
- **Undersized Terminal Connectors:** Terminal blocks cannot accept heavy-gauge wire necessary for high-current mains loads [1].
- **Rigid Supply Voltage Dependency:** Operating with a fixed 12V relay coil limits board compatibility when operating in 5V, 9V, or 24V control environments [1].
- **Fixed Capacitor Footprint Constraints:** Component pad spacing cannot accommodate larger electrolytic capacitors required for extended time delay ranges [1].

Our custom 4-layer KiCad design explicitly resolves these flaws by implementing wide trace geometry, thermal reliefs, high-voltage isolation slots, standard terminal spacing, and multi-rail compatibility.

6 Conclusion

The Timer Controlled Relay PCB project was successfully designed, routed, and verified using KiCad EDA. The board fulfills all design specifications by utilizing a multi-layer (4-layer) stackup, complete circuit protection elements, descending-voltage RC timing theory, and photorealistic 3D DRC verification.

References

- [1] FJRG76, "*On-Delay Timer with 555 IC*," March 2023. [Online]. Available: <http://fjrg76.com/2023/03/02/ondelay-timer-555-en/>. [Accessed: 30-Sep-2026].